GGG-A-450A
December 10, 1993
SUPERSEDING
GGG-A-00450 (GSA-FSS)
April 13, 1964

FEDERAL SPECIFICATION

ALIGNMENT TOOL, ELECTRONIC EQUIPMENT

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers alignment tools for adjustment of electronic circuits. The handles, or bodies, are made of nonconductive material so as to reduce to a minimum the change in radio frequency inductance or capacitance in the circuits under adjustment.

1.1.1 Federal specification coverage. This Federal specification does not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and sizes. The alignment tools covered by this specification shall be of the following types and sizes, as specified (see 6.1):

Type I	-	With one projecting metal tip
Size 6	-	Nominal length 6 inches (see figure 1)
Type II	-	With two metal tips, one recessed and one projecting
Size 2-1/2	-	Nominal length 6 inches (see figure 2)
Size 5	-	Nominal length 5 inches (see figure 3)
Type III	-	With two metal tips, both recessed
Size 7	-	Nominal length 7 inches (see figure 4)
Type IV	-	With one metal tip, recessed, and one metal or molded plastic slotted drive member
Size 6	-	Nominal length 6 inches (see figure 5)
Type V	-	With one projecting plastic tip and a plastic screwdriver-type handle
Size 7	-	Nominal length 7 inches (see figure 6)
Type VI	-	With two projecting plastic tips of different sizes
Size 7	-	Nominal length 7 inches (see figure 7)

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of alignment tools otherwise conforming to this specification.

3.2 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, molded, or indented

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directly on the item's surface in such a manner that it remains clear and legible throughout the life of the item.

3.3 Material. The handle or body of each alignment tool covered by this specification shall be made of a high grade nonconductive plastic so as to reduce to a minimum the change in radio frequency or capacitance in the circuits under adjustment. Except for types V and VI tools, each tip shall be made of stainless steel. Each type V and type VI tool shall be of a single piece of plastic, suitable for adjusting small screws.

3.4 Tip(s). The tip(s) of each alignment tool shall be of the flat-tip type for adjusting screws with a single slot. Each projecting metal tip shall be 1/8 inch in width, within a tolerance of ± 1/64 inch, and shall project from the handle or body 1/8 inch, within a tolerance of ± 1/32 inch. Each recessed metal tip shall either touch, or nearly touch, each inside wall of the cylinder in which it is recessed. Each recessed tip shall be at least 1/8 inch in length, within the hole in which it is recessed. The tip of each recessed tip shall be at least 1/32 inch and not more than 3/32 inch below the surface of the end of the handle or body. Each metal tip shall be 0.015 inch thick, within a tolerance of ± 0.002 inch. No metal tips or other metal members shall extend through the handle, or to such depth in the handle as to create a shock hazard. Plastic tips shall be as shown in the illustrations.

3.5 Slotted drive member. The slotted drive member of type IV tools shall be either flush with the end of the tool handle or recessed not more than 1/32 inch. The drive member shall be of a molded plastic or steel plug 0.175 inch ± 0.005 inch in diameter, with a slot 0.055 inch ± 0.002 inch in width cut across the end of the plug to a depth of at least 1/8 inch so as to provide a female drive.

3.6 Assembly. The metal tips and slotted drive members shall be firmly assembled in the tools. The tip(s) or drive slot shall be centrally located in the end of the tool.

3.7 Tolerances. Unless otherwise specified, a tolerance of ± 1/4 inch will be permitted on the overall length, and a tolerance of ± 1/64 inch shall be permitted on the outside diameter of all tools covered by this specification.

3.8 Types and sizes. Each type and size of alignment tool shall be of the applicable type and size shown in the illustrations (see figures 1 through 7), as specified. The alignment tools shall be of the dimensions specified herein.

3.9 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

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3.10 Workmanship. Workmanship shall be first class in every respect. The metal tip(s) shall be firmly held in the tool handle or shank(s). The tips shall have the burrs removed and shall be suitable for adjusting screws.

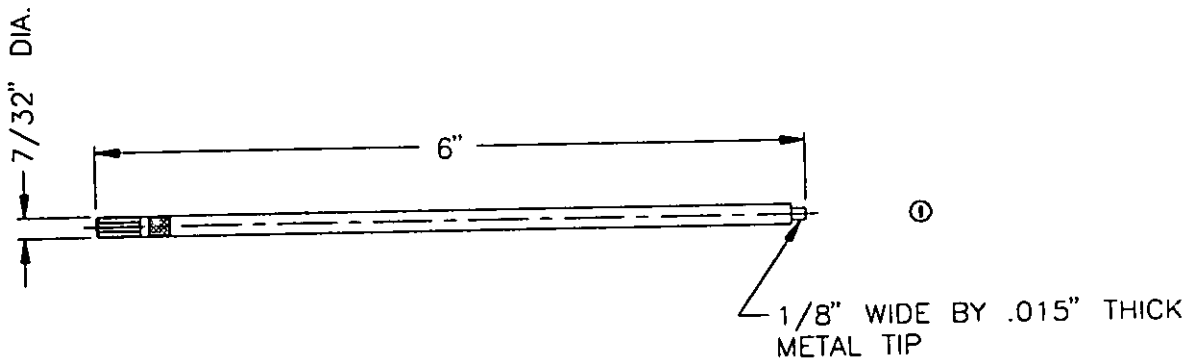


Figure 1. Type I, size 6, alignment tool with one projecting metal tip (see 3.7).

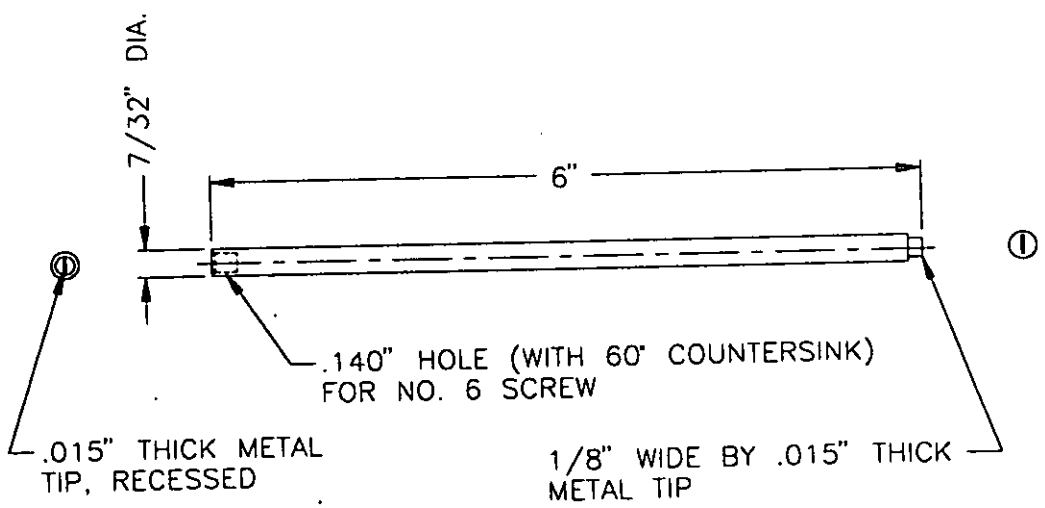


Figure 2. Type II, size 2-1/2 alignment tool with two metal tips, one recessed and one projecting (see 3.7).

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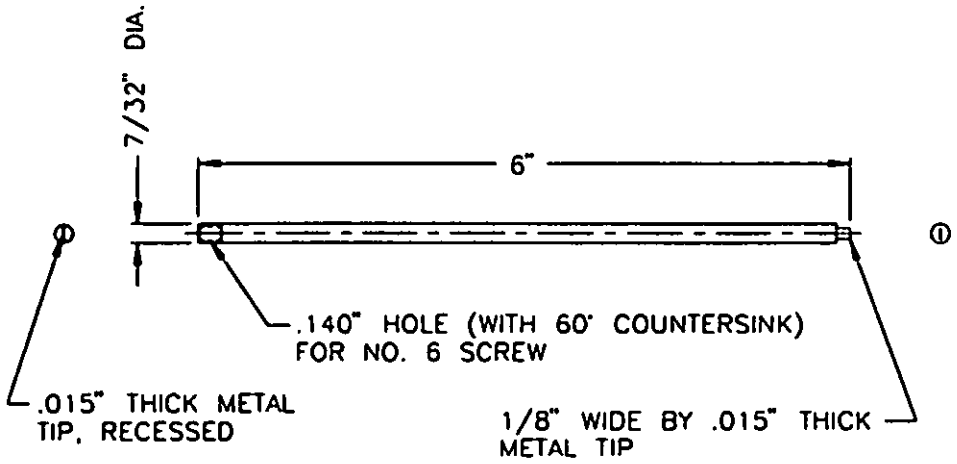


Figure 3. Type II, size 5, alignment tool with two metal tips, one recessed and one projecting (see 3.7).

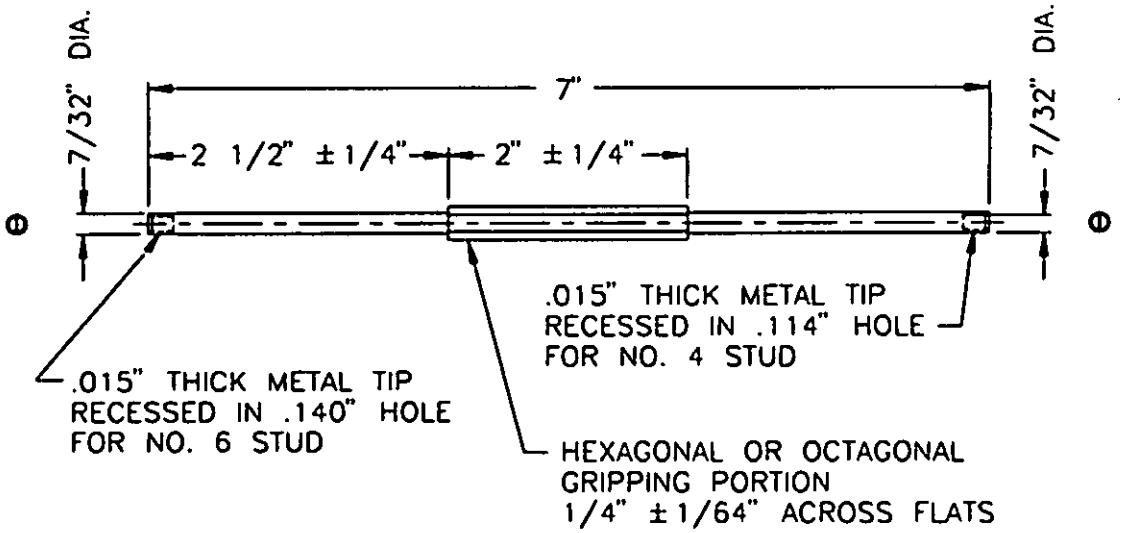


Figure 4. Type III, size 7, alignment tool with two metal tips, both recessed (see 3.7).

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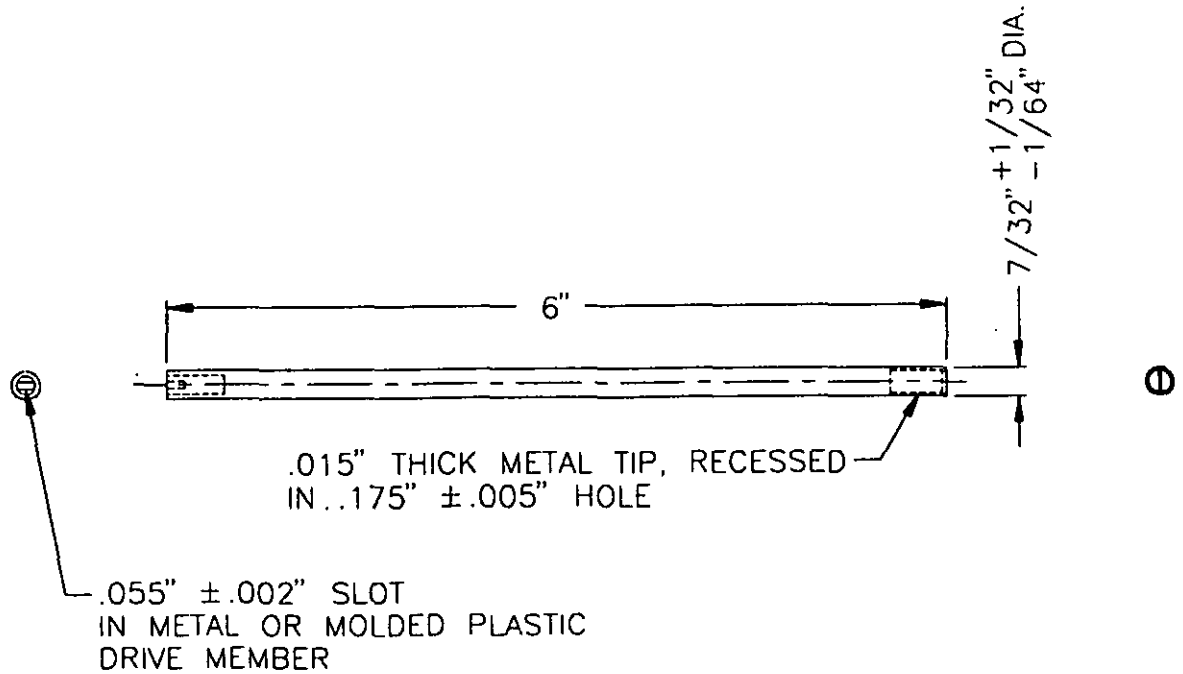


Figure 5. Type IV, size 6, alignment tool, with one recessed metal tip, and one metal, or molded plastic drive member (3.7).

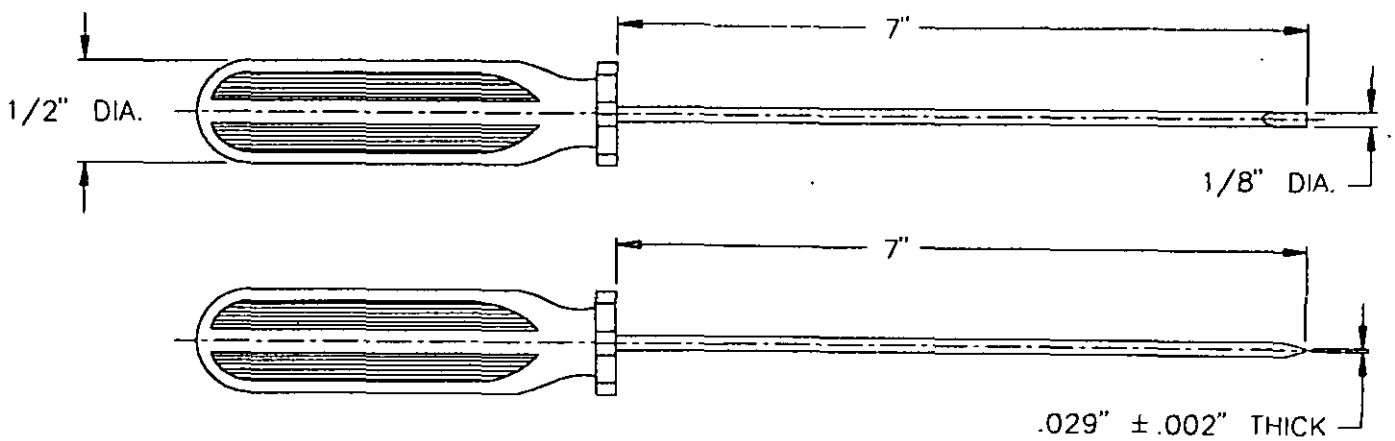


Figure 6. Type V, size 7, alignment tool with one projecting plastic tip with yellow amber handle (see 3.7).

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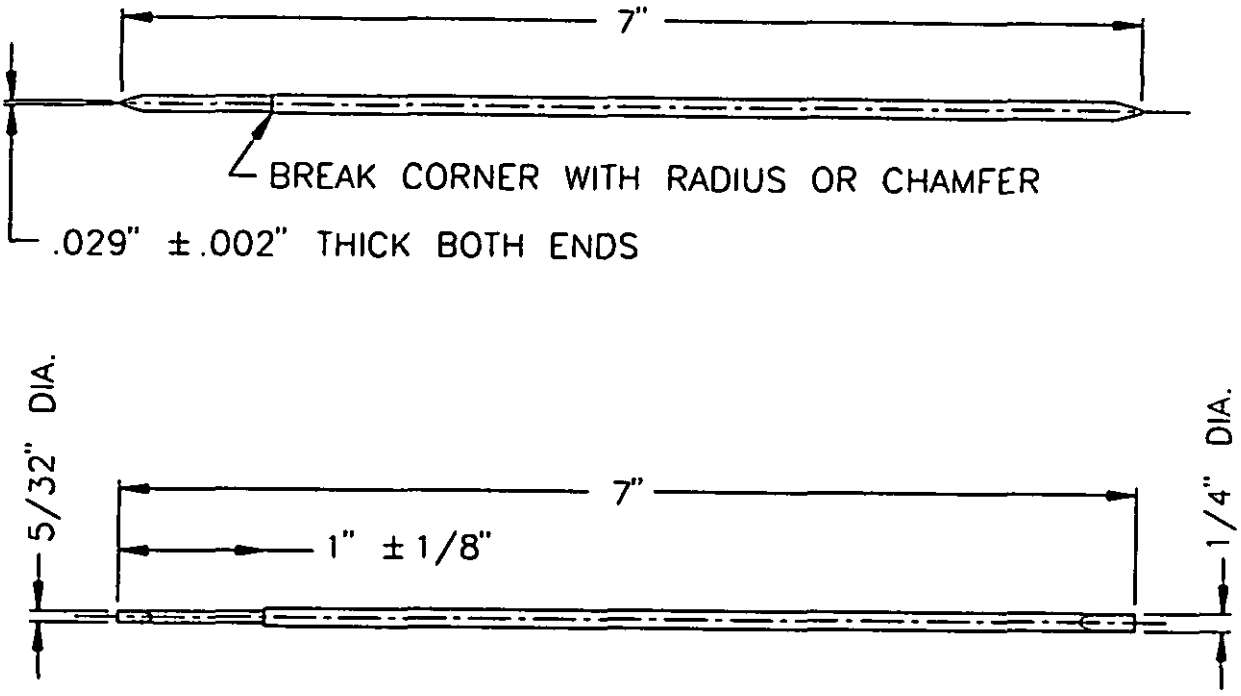


Figure 7. Type VI, size 7, alignment tool with two projecting plastic tips of different sizes (see 3.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Place of inspection. Inspection shall be performed at the place of manufacture.

4.3 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.3.1 Lot. All alignment tools of the same type and size, offered for delivery at one time, shall be considered a lot for purposes of sampling and inspection.

4.3.2 Visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in terms of alignment tools. The sample unit shall be one alignment tool. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of percent defective, shall be 2.5.

TABLE I. End item visual defects

Examine	Defect
Construction and workmanship	Material not as specified
	Tips not size or thickness specified
	Not assembled as specified
	Fractured, broken, cracked, or malformed
	Sharp or rough edges, burrs, or projections
Marking	Missing, illegible, incorrect, or incomplete
	Not as specified

4.3.3 Dimensional examination. The end items shall be examined for any nonconformance with dimensional requirements. The lot size shall be expressed in terms of alignment tools. The sample unit shall be one alignment tool. The inspection level shall be II and the AQL, expressed in terms of percent defective, shall be 2.5.

4.4 Examination. Each of the sample alignment tools selected in accordance with 4.3.2 and 4.3.3 shall be examined to verify conformance with all the requirements of this specification. Any tool in the sample containing one or more defects shall be rejected and if the number of defective tools in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 and as specified in table II. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE II. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and size of item required (see 1.2.1).
c. Level of preservation, packaging, and packing required (see 5.1).

Custodians:

Army - GL
Air Force - 99

Review Activity:

Air Force - 84

MILITARY COORDINATING ACTIVITY:

Army - GL

PREPARING ACTIVITY:

GSA - FSS

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